

Solution Code - Module 3-Core Java - SimpleCalculator

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q02_SimpleCalculator.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q02_SimpleCalculator {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.println("=== Simple Calculator ===");
7 |
8 |         try {
9 |             System.out.print("Enter first number: ");
10 |             double num1 = scanner.nextDouble();
11 |
12 |             System.out.print("Enter second number: ");
13 |             double num2 = scanner.nextDouble();
14 |
15 |             System.out.print("Choose an operation (+, -, *, /): ");
16 |             char operation = scanner.next().charAt(0);
17 |
18 |             double result = 0;
19 |             boolean validOperation = true;
20 |
21 |             switch (operation) {
22 |                 case '+':
23 |                     result = num1 + num2;
24 |                     break;
25 |                 case '-':
26 |                     result = num1 - num2;
27 |                     break;
28 |                 case '*':
29 |                     result = num1 * num2;
30 |                     break;
31 |                 case '/':
32 |                     if (num2 == 0) {
33 |                         System.out.println("Error: Division by zero is not allowed.");
34 |                         validOperation = false;
35 |                     } else {
36 |                         result = num1 / num2;
37 |                     }
38 |                     break;
39 |                 default:
40 |                     System.out.println("Error: Invalid operation chosen.");
41 |                     validOperation = false;
42 |             }
43 |
```

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```
44 |         if (validOperation) {
45 |             System.out.printf("Result: %.2f %c %.2f = %.2f\n", num1, operation, num2, result);
46 |         }
47 |     } catch (Exception e) {
48 |         System.out.println("Error: Invalid input. Please enter numbers.");
49 |     } finally {
50 |         scanner.close();
51 |     }
52 | }
53 | }
```